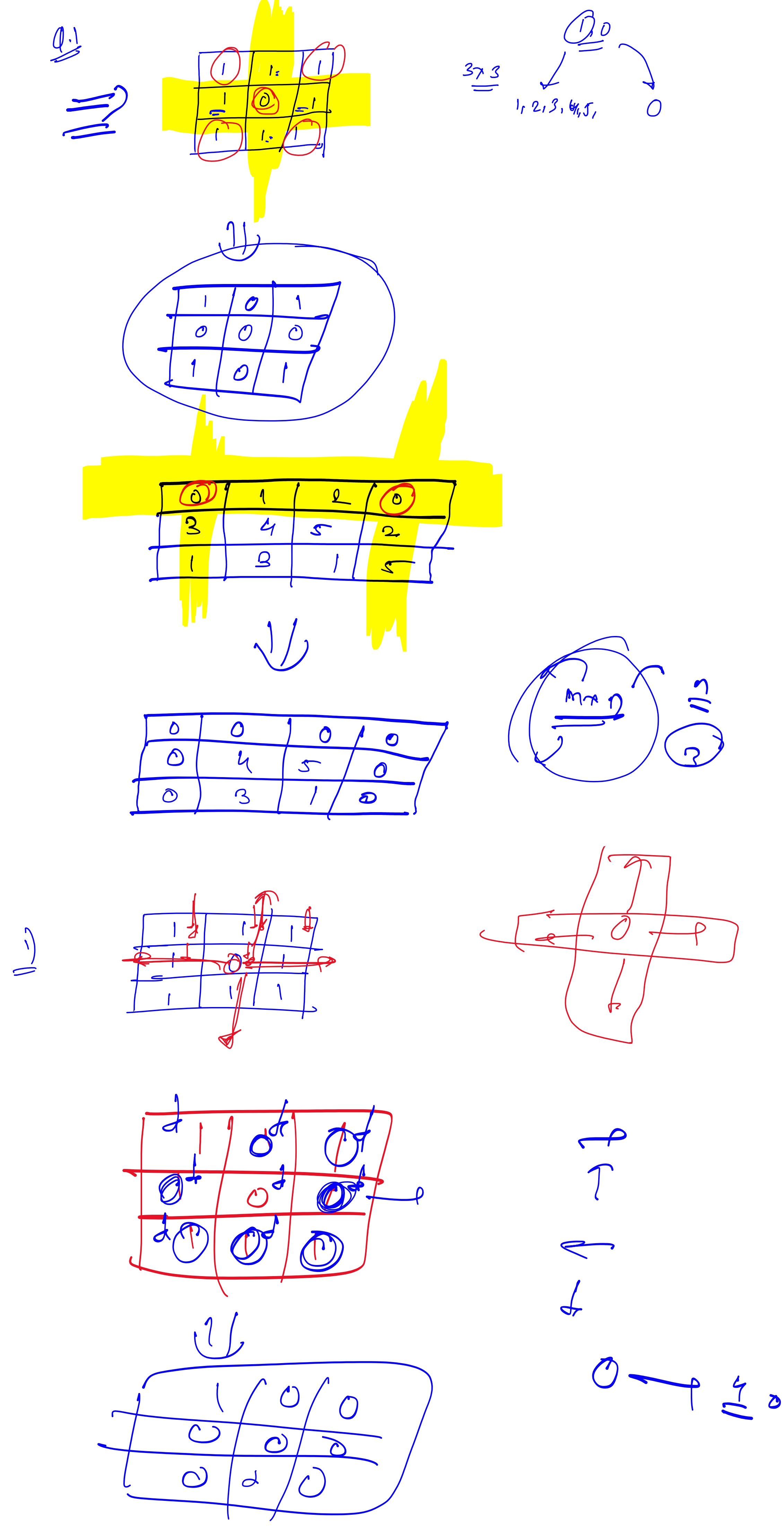
Arrays - XII

Tuesday, 26 May 2026 8:39 PM

Q.1

2D / matrix Highest Probability



Approach -

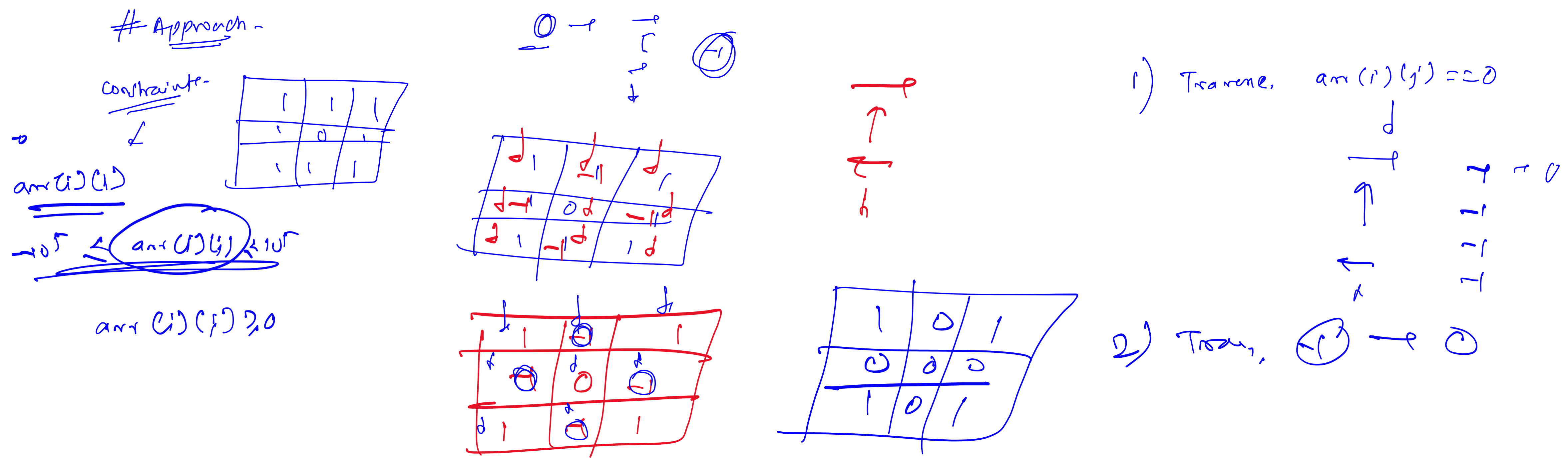
Constraints -

$-10^5 \le arr[i][j] \le 10^5$

arr[i][j] ≥ 0

1) Traverse, arr[i][j] == 0

2) Traverse, -1 → 0



Code -

```
for(int i=0; i< arr.length; i++){
    for(int j=0; j< arr[0].length; j++){
        if (arr[i][j] == 0){
            for(int k=0; k< arr[0].length; k++){
                if (arr[i][k] != 0){
                    arr[i][k] = -1;
                }
            }

            for(int k=0; k< arr.length; k++){
                if (arr[k][j] != 0){
                    arr[k][j] = -1;
                }
            }
        }
    }
}
```

// Step 1

// TC - $(m \times n \times (m+n)) + (m \times n)$

$mn \times (m+n) + mn$

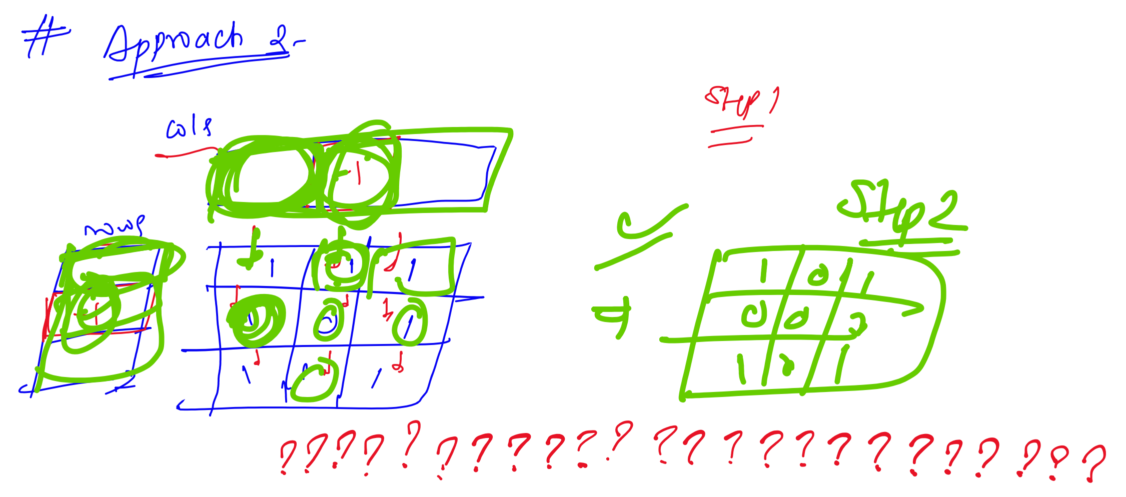
$m^2n + mn^2 + mn$

// SC - O(1)

```
for(int i=0; i< arr.length; i++){
    for(int j=0; j< arr[0].length; j++){
        if (arr[i][j] == -1){
            arr[i][j] = 0;
        }
    }
}
```

// Step 2

Approach 2 -



??????????????????????????

Code -

```
int[] rows = new int[arr.length];
int[] cols = new int[arr[0].length];
```

```
for(int i=0; i< arr.length; i++){
    for(int j=0; j< arr[0].length; j++){
        if (arr[i][j] == 0){
            rows[i] = -1;
            cols[j] = -1;
        }
    }
}
```

// Step 1

// TC - O(m×n)

// SC - O(m + n)

```
for (int i=0; i< arr.length; i++){
    for(int j=0; j< arr[0].length; j++){
        if ( rows[i] == -1 || cols[j] == -1){
            arr[i][j] = 0;
        }
    }
}
```

// Step 2

Approach 3 -